

Quick Check

Lesson 5: Sensemaking: Connecting the Chemistry of Food to Health Outcomes in Kisumu

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 5

Two children in a village in Kisumu eat the same amount of food each day. One eats only ugali (maize starch). The other eats githeri (maize and beans). Based on food test results, what is the most important chemical difference between ugali and githeri?

- A** Githeri contains protein as well as carbohydrate, but ugali is mainly carbohydrate
- B** Ugali contains protein, but githeri does not
- C** Ugali contains more food than githeri, so it gives less energy
- D** Githeri contains no carbohydrates, but ugali does

In a Biuret test, a bean extract turns purple and a sample of ugali stays blue. What does this result show?

- A** Beans contain starch and ugali does not
- B** Ugali contains more lipids than beans
- C** Beans contain vitamin C and ugali does not
- D** Beans contain protein, while ugali has little or no protein

In a diet audit, Child 1 eats ugali three times a day. Child 2 eats githeri twice and ugali once. Which statement about their diets is correct?

- A** Child 2's diet supplies protein but no carbohydrates
- B** Child 1's diet supplies carbohydrates but too little protein, and Child 2's diet supplies both
- C** Child 1's diet has no carbohydrates, but Child 2's diet has carbohydrates
- D** Both diets supply the same chemicals of life because they contain the same amount of food

A learner tests orange juice with DCPIP in the lab and it changes colour. A disease is caused by a lack of this same chemical in the diet. Which disease is it?

A Marasmus

B Kwashiorkor

C Scurvy

D Rickets

A child with kwashiorkor has a swollen belly, thin limbs and discoloured hair. Which row correctly links a symptom to the protein that is lacking?

A Swollen belly: keratin loss

B Thin limbs: keratin loss

C Swollen belly: albumin loss

D Discoloured hair: albumin loss

Which statement is the best scientific explanation for why the child who eats only ugali develops kwashiorkor, while the child who eats githeri stays healthy?

- A** The ugali child does not get enough energy because ugali has no carbohydrates
- B** The ugali child gets enough energy from starch but too little protein for building body tissues
- C** The githeri child gets more vitamins, and vitamins alone build muscle and hair
- D** The ugali child eats a smaller quantity of food than the githeri child

A group is filling in the protein box of their 'Our Chemicals of Life Model' template. Which entry is correct?

A Food source: orange juice; consequence of deficiency: rickets

B Food source: beans; consequence of deficiency: kwashiorkor

C Food source: ugali; consequence of deficiency: scurvy

D Food source: ugali; consequence of deficiency: anaemia

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).